The Hang Seng University of Hong Kong
Department of Computing
COM6001– Computing with Big Data
Group Assignment Specification – Data Visualization and Analysis

This group-level assignment constitutes 30% of your overall course grade. You should form a group of **at most 3 members**. All the group members are required to contribute actively and significantly to the assignment. **Your role allocation needs to be included in the final report**.

Context

You are going to perform a data analysis and visualization on Hong Kong's transportation based on the mandatory and elective datasets.

Deliverables

1. Online form to state your datasets and techniques used – this is for cross-module self-plagiarism prevention (**individual submission** via this MS forms).
2. Upload at least three papers that you read and download from IEEE Xplore (group submission via Moodle)
3. Your **printed text-rich** jupyter notebook with all code, plots, insights and discussion well presented, in pdf format (group submission via Moodle)
4. A zipped folder that contains (group submission via Moodle)
   a. your Jupyter notebook in *.ipynb* format
   b. your datasets used in a sub-folder. Make sure your notebook can load all needed datasets with the sub-folder setting
   c. if your zip file is too big, please submit a link in Moodle, and also send me your link via email

Presentations will be based on the **handed-in version** of the notebook. You can use either the printed pdf or the jupyter notebook to present, so that animated plots (if any) can be demonstrated.

Important dates

Project submission: 30 November 2025, Sunday (11:59 pm)
Project presentation: 1 Dec (18:30 – 21:30), 2 Dec (15:00 – 18:00) and 3Dec (16:00 – 19:00), with at most 10 groups presenting in each day.

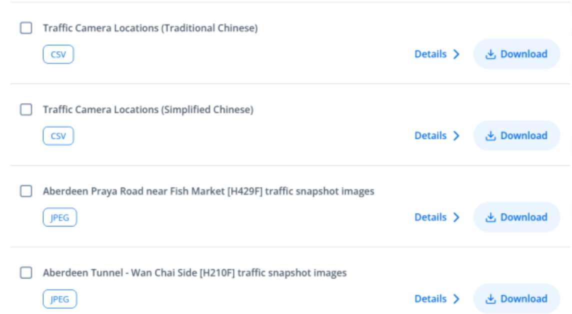
Recommended datasets

a) The main datasets are a collection of data files at [Road Traffic Accident Statistics]() that were collected and processed for 2017-2023. In each year, there are 45 unique csv files, presenting various statistics over local accidents. Therefore, in total, we have more 300 data files. **You must use these datasets for more than one year** to achieve some cross-year analysis on traffic accidents or casualties over factors like regions, type of vehicle, type of road users, pedestrian locations, casualty ages, accident severity and so on. Note: not all 45 csv files need to be used in your analysis, just pick the ones that you think suitable and insightful.

Mandatory datasets:		
M1. **Road Traffic Accident Statistics** (Year 2017-2023) Search by the string "Road Traffic Accident Statistics Year" at data.gov.hk or go to this link.	Year by year data about accidents (number of casualties, severity, types of the involved vehicles, types of people etc.). These datasets may be used to visualize the accident trends over the years, age, vehicle types, severity levels, road user types and locations.	45 csv files in each year for 7 years

b) In addition to the data from a), you should also consider **at least one dataset** from the following recommended. You can also include **any** other Hong Kong related datasets to help you get more insightful observations and analysis. A list of recommended datasets include but not limited to:

Elective datasets:		
E1. **Special Traffic News (2nd Generation)**	This dataset tells you about the details about traffic incidents in a real-time and event-by-event format. The updates about the same incident will have the same INCIDENT-NUMBER. Below shows one message about incident IN-25-05498 which occurred on 2 Sept, 2025. https://data.gov.hk/en-data/dataset/hk-td-tis_19-special-traffic-news-v2 ▼<message> <INCIDENT_NUMBER>IN-25-05498</INCIDENT_NUMBER> <INCIDENT_HEADING_EN>Road Incident</INCIDENT_HEADING_EN> <INCIDENT_HEADING_CN>道路事故</INCIDENT_HEADING_CN> <INCIDENT_DETAIL_EN>Traffic Accident</INCIDENT_DETAIL_EN> <INCIDENT_DETAIL_CN>交通意外</INCIDENT_DETAIL_CN> <LOCATION_EN>Clear Water Bay Road</LOCATION_EN> <LOCATION_CN>清水灣道</LOCATION_CN> <DISTRICT_EN/> <DISTRICT_CN/> <DIRECTION_EN>Sai Kung</DIRECTION_EN> <DIRECTION_CN>西貢</DIRECTION_CN> <ANNOUNCEMENT_DATE>2025-09-02T14:20:00</ANNOUNCEMENT_DATE> <INCIDENT_STATUS_EN>CLOSED</INCIDENT_STATUS_EN> <INCIDENT_STATUS_CN>完結</INCIDENT_STATUS_CN> <NEAR_LANDMARK_EN>Ngau Chi Wan Sports Centre</NEAR_LANDMARK_EN> <NEAR_LANDMARK_CN>牛池灣體育館</NEAR_LANDMARK_CN> <BETWEEN_LANDMARK_EN/> <BETWEEN_LANDMARK_CN/> <ID>121734</ID> <CONTENT_EN>All lanes of Clear Water Bay Road (Sai Kung bound) near Ngau Chi <CONTENT_CN>較早前因交通意外而封閉的清水灣道(往西貢方向)近牛池灣體育館的全線現已解 <LATITUDE/> <LONGITUDE/> With this real-time, case by case data, you may be able to conduct analysis and draw insights such as "which road segments are more likely to have accident", "how the traffic accidents are related to time and location", "how long a traffic accident can be resolved" and etc. You need to avoid to count the same accident repeatedly.	Each day including about 100+ messages in **xml** format

<table>
<tr>
<td>E2.
Real time CCTV snapshots</td>
<td>This dataset tells you about the latest CCTV images (real time, JPEG) and CCTV locations (CSV) in Hong Kong.
https://data.gov.hk/en-data/dataset/hk-td-tis_2-traffic-snapshot-images


The CCTV images may be used to visualize the current traffic condition in a specific location. The CCTV locations may be used with dataset E.1 to obtain the geolocations (the longitude and altitude values) of the accident.</td>
<td>JPEG, csv</td>
</tr>
<tr>
<td>E3.
Locations and measurements of Traffic Detectors</td>
<td>https://data.gov.hk/en-data/dataset/hk-td-sm_4-traffic-data-strategic-major-roads
This includes several dataset as shown below.


Traffic Speed, Volume and Road Occupancy (Raw Data).xml tells you about the measurement results by the detectors. The information elements are as follows.

This XML file does not appear to have any style infor
<raw_speed_volume_list xmlns:xsi="http://www.w3.o
<date>2025-09-26</date>
<periods>
<period>
<period_from>16:10:00</period_from>
<period_to>16:10:30</period_to>
<detectors>
<detector>
<detector_id>AID01101</detector_id>
<direction>South East</direction>
<lanes>
<lane>
<lane_id>Fast Lane</lane_id>
<speed>104</speed>
<occupancy>0</occupancy>
<volume>2</volume>
<s.d.>14.8</s.d.>
<valid>Y</valid>
</lane>
<lane>
<lane_id>Middle Lane</lane_id>
This data can be used for analysis towards questions like “how vehicles speed changes over time on a road?”, “how the road speed and volume are related to the accident chances?”
Locations of Traffic Detectors.csv tells you about the geolocations of the detectors, which may be used to plot the locations on a map.</td>
<td>csv, xml</td>
</tr>
<tr>
<td>E4.
Car Journey Time data</td>
<td>https://data.gov.hk/en-data/dataset/hk-td-tis_34-car-journey-time-data
Peak hour average journey time of selected routes in Hong Kong for 2023 and 2024.
A.M. Peak Car Journey Time – 2023 or 2024
P.M. Peak Car Journey Time – 2023 or 2024
A.M. Peak Journey Time – 2023 or 2024
P.M. Peak Journey Time – 2023 or 2024</td>
<td>Eight excel files</td>
</tr>
</table>

E5. Tropical cyclone best track data (post analysis)	Provides data on post analysed position and intensity of tropical cyclones https://data.gov.hk/en-data/dataset/hk-hko-rss-tropical-cyclone-best-track-data	csv

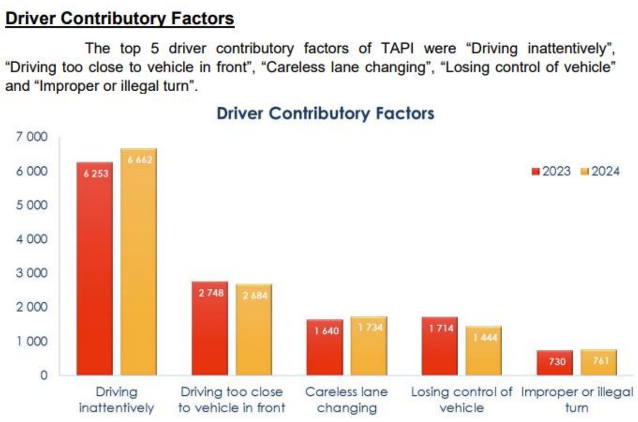
Objectives and sample analyses:

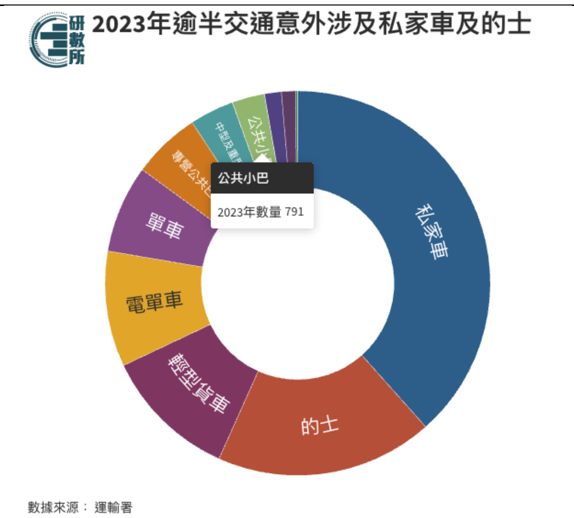
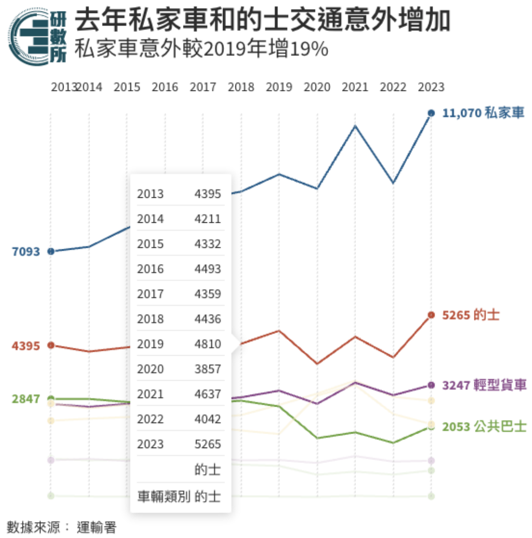
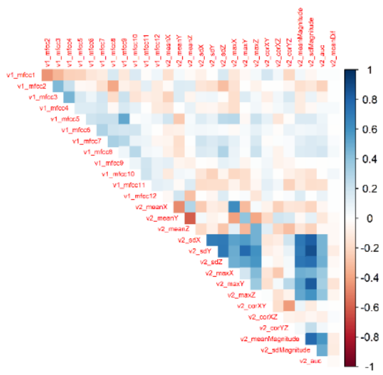
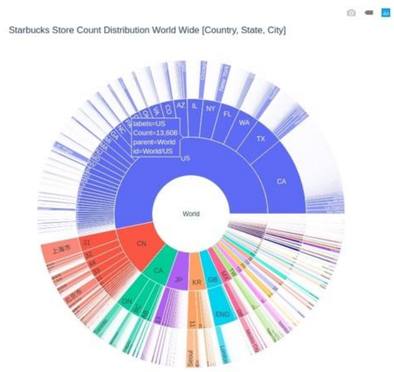
Your analysis should aim for one or more of the following:

1) To summarize and visualize accident aggregative statistics in Hong Kong (from dataset M1)
2) To find out interesting insights from the traffic patterns, e.g., “it is more likely to have congestion if you travel from New Territories to Kowloon but not the opposite direction” or “some roads are having high frequency of speeding during the mid-night hours” (using E3/E4 and other datasets)
3) To find out impacting factors of traffic accidents, such like bad weather (rain or windy, peak or off-peak hours, locations like traffic black spots. (using E1/E2/E5 and other datasets)
4) To describe/predict the resolution time (how long the accident can be resolved) of different types of accidents with other features like weather, time of the day, location etc. (using dataset E1)

Sample works from HK Police and HK01 listed below are for your reference and inspirations only. You may replicate similar visualizations using the raw datasets and Python in your report, where you should provide the datasets, your own code, the code explanation as well as your own thoughts on the message conveyed by the plots. **More importantly, you should do better than just tables and bar charts!!**

[1]. Hong Kong Police, “*Traffic report 2024,*” https://www.police.gov.hk/info/doc/statistics/traffic_report_2024_en.pdf.
[2]. Hong Kong 01 News, 《交通意外｜10年升過千宗　嚴重意外減6成　最多涉及私家車和的士》，https://www.hk01.com/article/60211448?utm_source=01articlecopy&utm_medium=referral

	A basic bar chart. Source: HK Police’s report

<table>
<tr>
<td>
</td>
<td>An interactive pie chart. Source: HK01</td>
</tr>
<tr>
<td>
</td>
<td>An interactive line chart. Source: HK01</td>
</tr>
<tr>
<td> 
</td>
<td>More interactive charts can be used to show inter-feature correlations or statistics over time/location (these charts were not plotted using the traffic data)</td>
</tr>
</table>

Suggested content for your data analysis notebook

1. Introduction and literature study: Describe the data analysis project overall. Report your findings about the available datasets. Conduct a literature study on some existing research works. Search papers with relevant key words such as *"traffic accidents prediction machine learning"* on IEEE Xplore from a **campus PC or through campus WiFi,** read at least **three papers** and upload the papers that you read to Moodle. Include these papers in your references section of the report.

2. Data importing and merging: you need to merge and sometimes transform multiple datasets to form a more comprehensive dataframe for visualization and analysis.

3. Data cleaning: check for unnecessary features, deal with the missing values and the outliers, and deal with the inconsistencies in data.

4. Data analysis and visualization: summarize the main characteristics of the data using descriptive statistics. Visualize the data with a number of plots and explain the messages or insights found. Approaches may include, but not limited to: 1) summarize and describe Hong Kong general traffic and/or traffic accident patterns, 2) use aggregative statistics to show different patterns in different feature groups. 3) to conduct a time series visualization to show any trends over the days, months, seasons or years; and 4) to find and present any interesting relationships among various datasets (such as between weather, time/season, accidents, traffic conditions).

5. Conclusion and reflective journal: summarize your findings and insights from the data and present what you learn, and what were the challenges you overcame in doing this assignment.

6. **Role allocation: describe your job allocation in detail.**

Please note that the above contents are suggested contents only. Your group is free to propose your own content when appropriate. The point is that it is a **report-like ipynb notebook and will be used for your presentation**. You need to write down your thoughts, methods, motivations, and findings along with the codes.

Assessment criteria (Full mark: 30):

Assessment items	**Description**	**Marks**
Introduction and conclusion	Logic and flow of the introduction and conclusions. Quality of literature study.	3
Data importing and merging	Number of datasets used; techniques used in merging the datasets	4
Data cleaning and manipulation	Skills used in data manipulation, cleaning, and feature transforming. Please avoid hard-coding as much as possible.	5
Data analysis and visualization	Variety and comprehensiveness of the visualization tools used. Insightfulness of the data analysis.	8
Organization	Use a proper report structure (with proper cover page, headings, references, *etc.*) **List out all references**. Please obey to the University's guideline for using Generative AI Tools here. *Warning: if you do not provide any references, you may get zero marks in the organization part!	4
Presentation	How attractive is your presentation? Have you used effective presentation techniques instead of reading all the texts on the report? Can you explain the technical details and answer the teacher's questions?	6